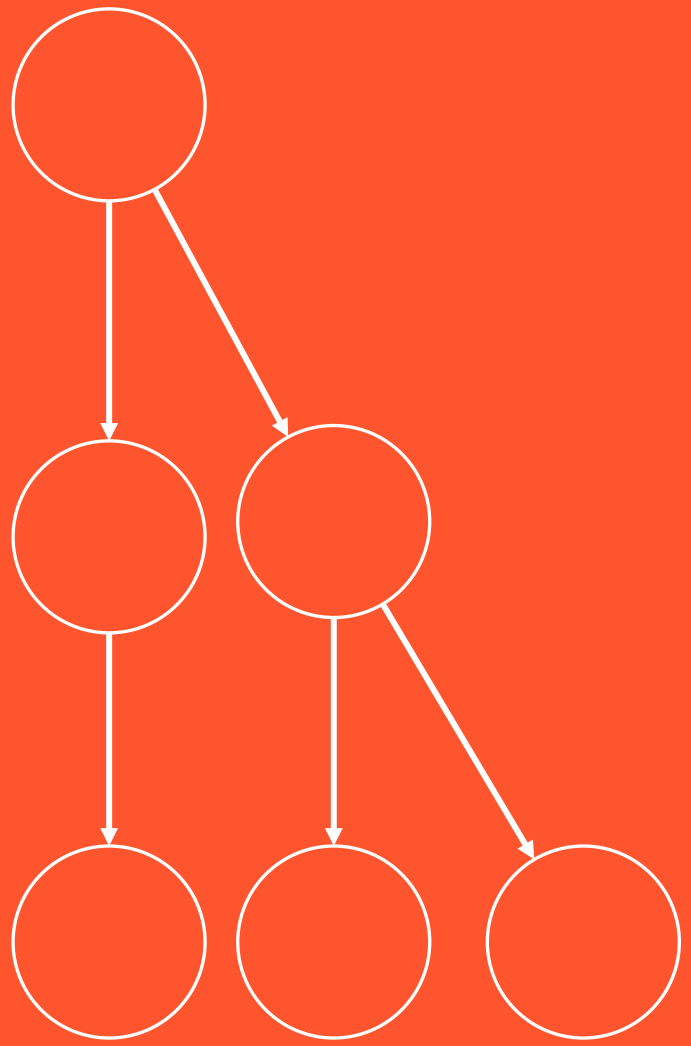


BCOG 100

Neural Circuits, Valence, and Affect

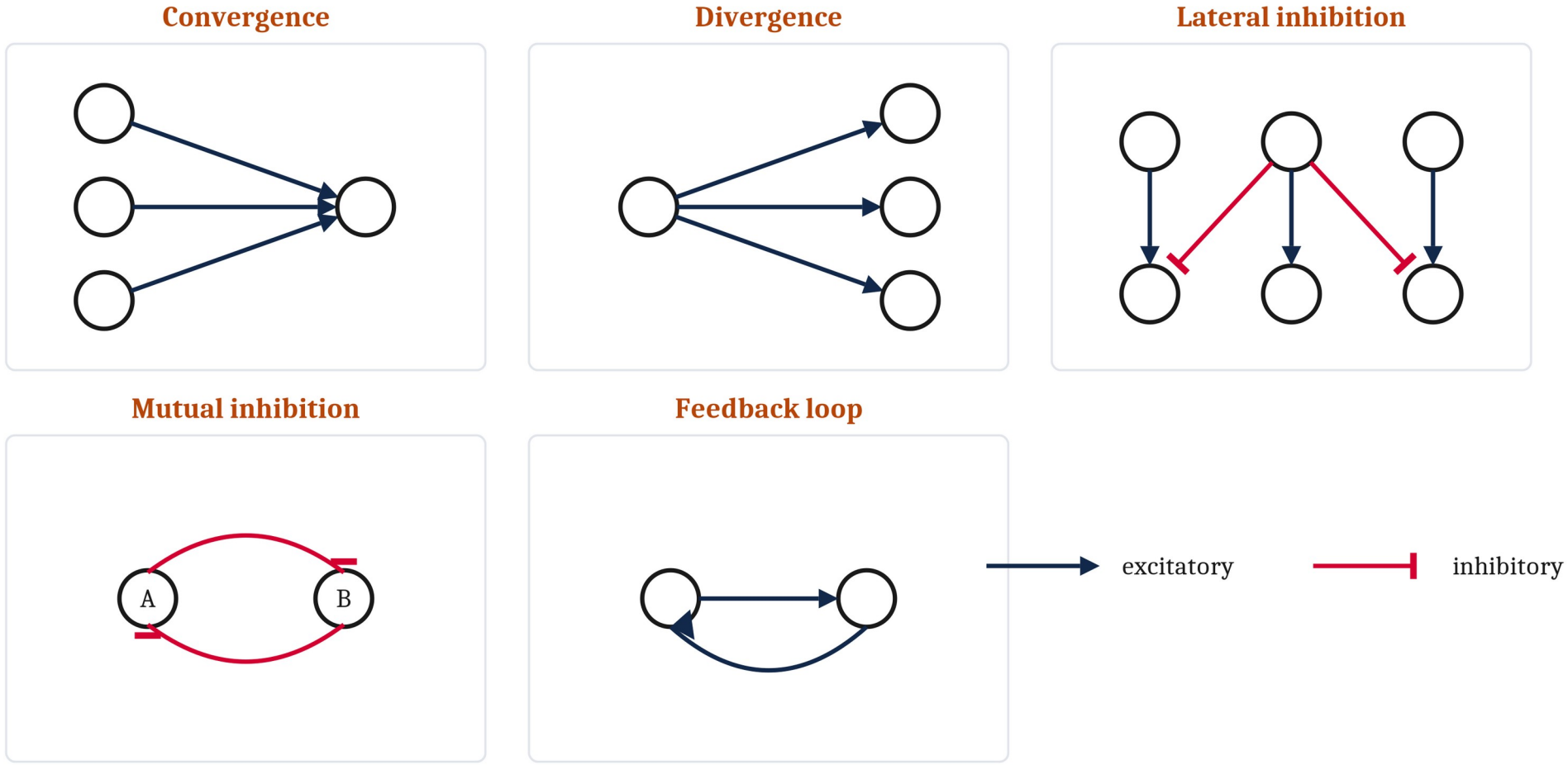
Lecture 4.2 — Circuits That Compute Value

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

From a Unit to a Wiring Diagram

Connection is a matter of **wiring**, not of proximity — two cells pressed together may have nothing to do with each other. So a nervous system can be **drawn**, and the drawing reasoned about without knowing anything about the cells.

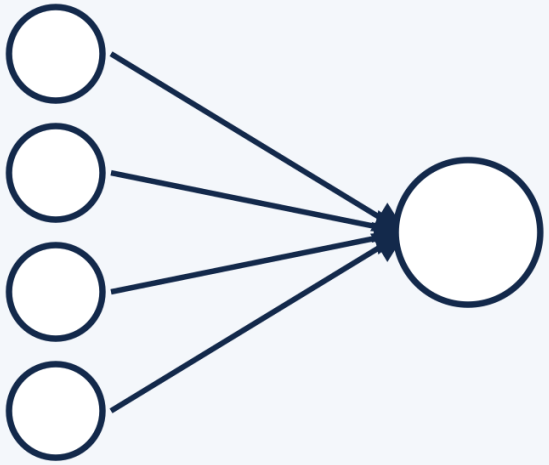
Five wiring patterns, and between them they do the work



The Two Simplest Motifs

Convergence — many cells onto one — is what makes combining possible, because a cell can only weigh what actually arrives at it. **Divergence** — one cell onto many — is what makes an action a whole-body commitment rather than a local twitch.

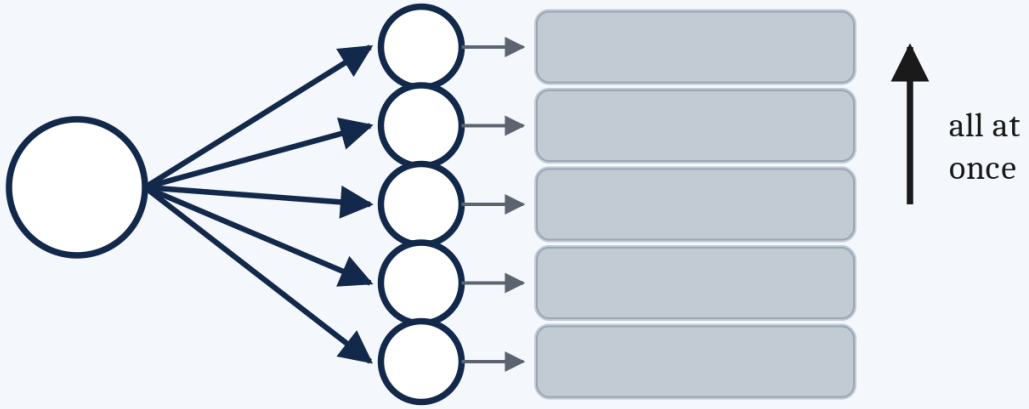
convergence



a cell can only weigh what actually arrives at it

the demand:
COMBINE

divergence



one verdict, reaching muscle all over the body

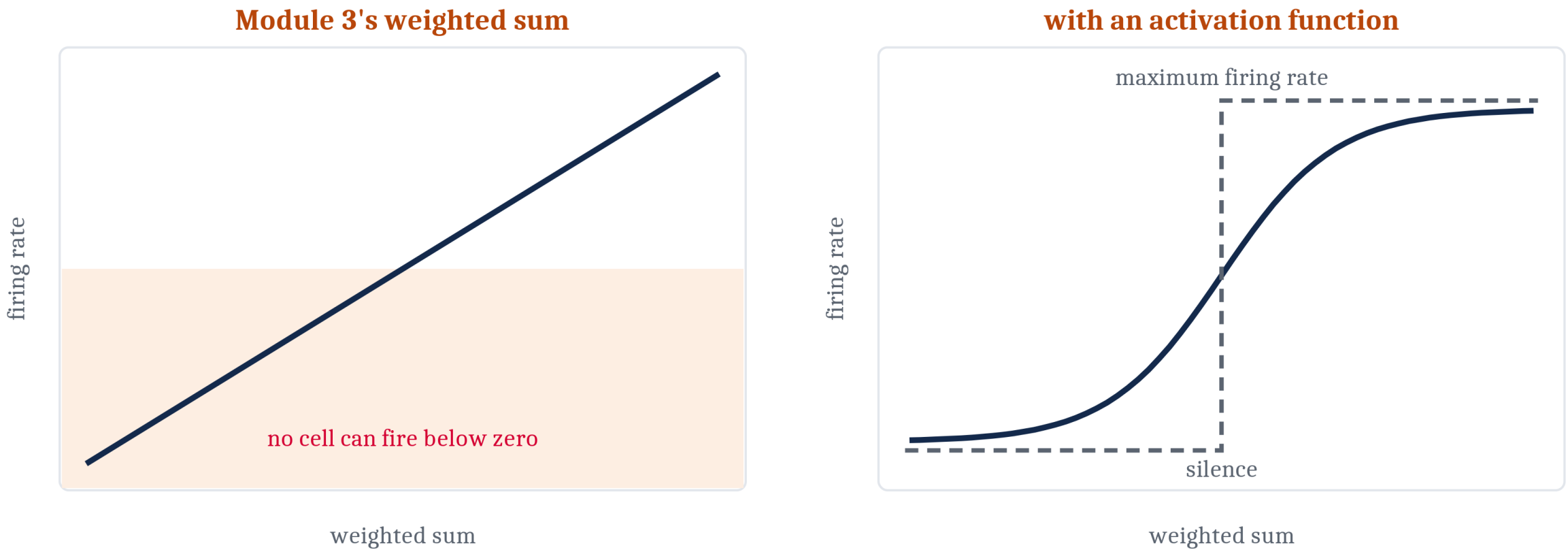
the demand:
COMMIT

Many onto one makes combining possible. One onto many makes an action a whole-body commitment.

The Floor and the Ceiling

A cell cannot fire less often than never, and cannot fire faster than its refractory period allows. So the weighted sum cannot be the final answer: it passes through an **activation function** — a hard **threshold function**, or a smooth **sigmoid function**.

Adding a floor and a ceiling



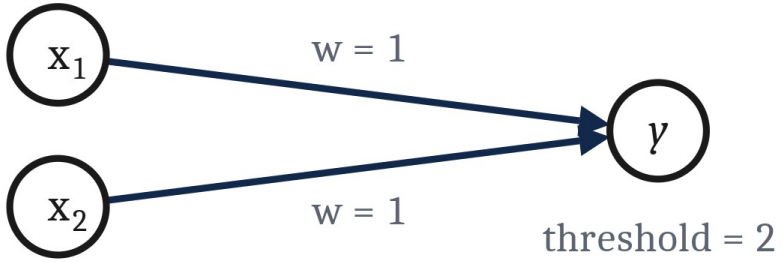
A hard threshold and a smooth S-shaped curve solve one problem two ways.

Two Numbers Make a Logic Gate

Two excitatory inputs, each with a weight of 1. Threshold **2** and the unit computes *AND*. Threshold **1** and it computes *OR*. **Nothing about the wiring changed** — only the threshold.

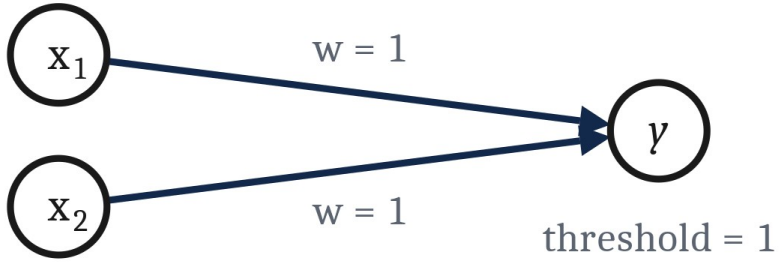
Identical wiring. One number changed.

AND



x_1	x_2	y
0	0	0
1	0	0
0	1	0
1	1	**1**

OR

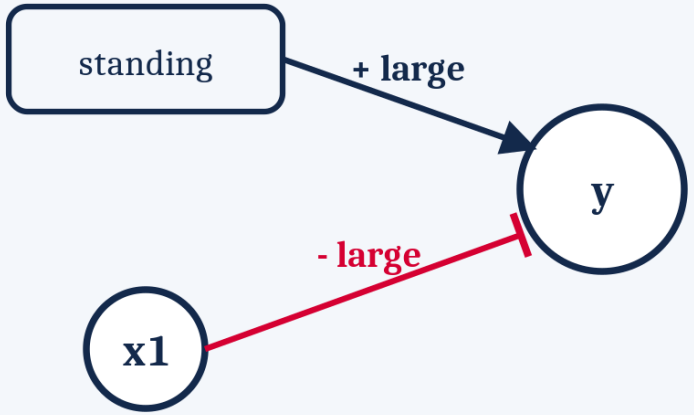


x_1	x_2	y
0	0	0
1	0	**1**
0	1	**1**
1	1	**1**

This, and Not That

An inhibitory input is a connection with a **negative weight**. Standing excitation cancelled by one gives *NOT*; wired alongside an excitatory input it gives *AND NOT*. The unit is not choosing, and does not know what *AND* means.

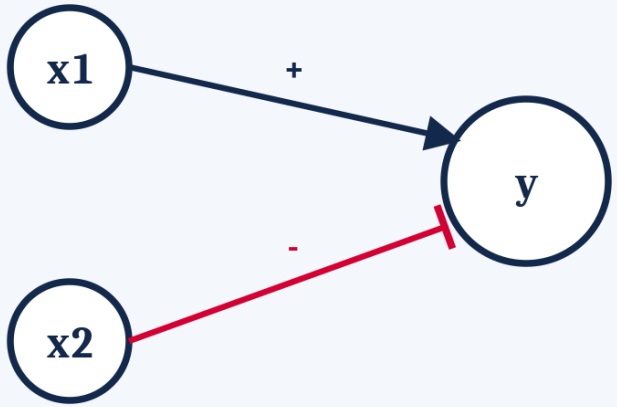
NOT



x1 silent -> y fires

x1 firing -> y silent

AND NOT



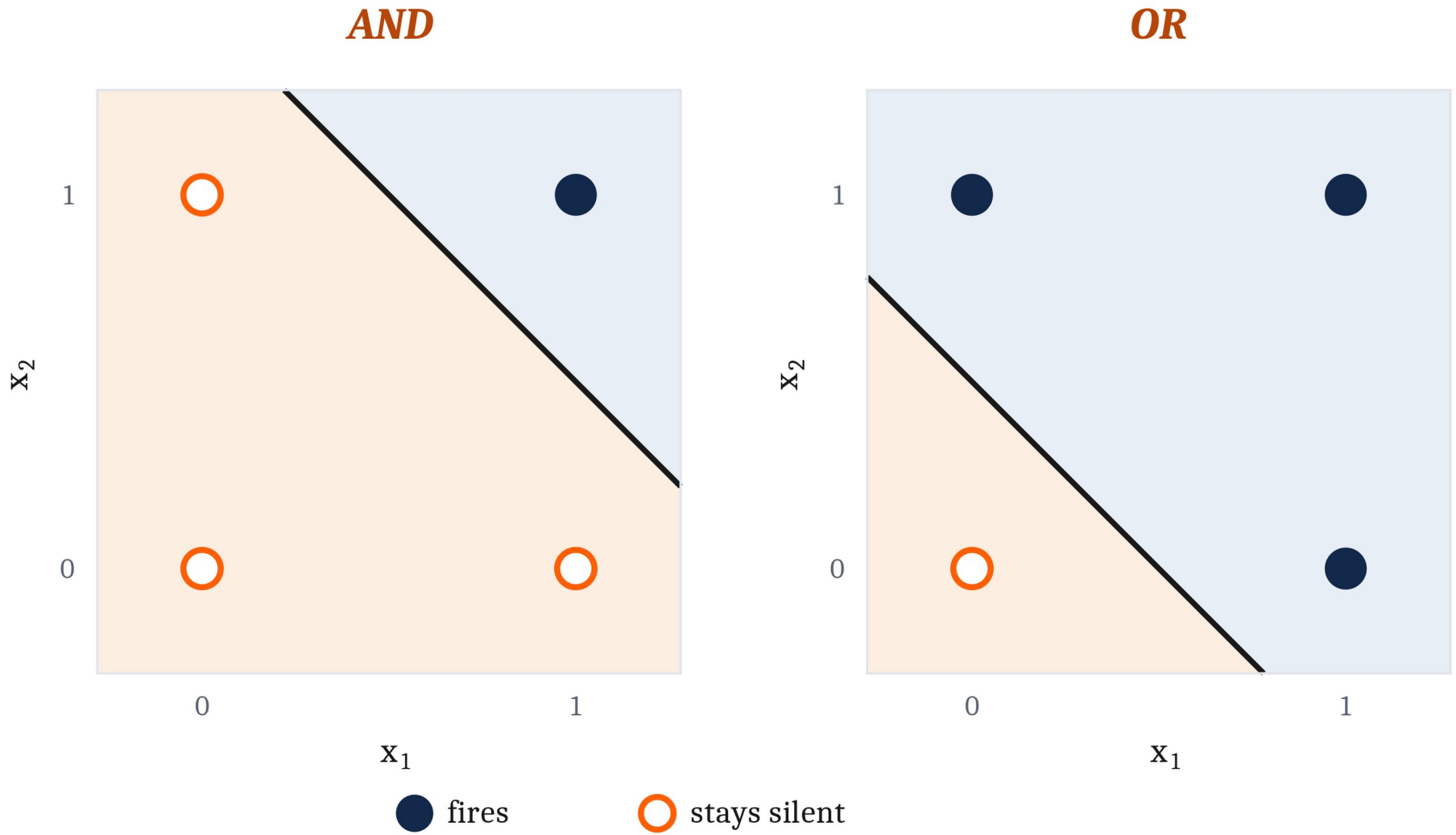
fires when x1 is present

and x2 is absent

The logic is in the weights and the threshold, and nowhere else. The unit is not choosing.

What a Unit Computes, as a Picture

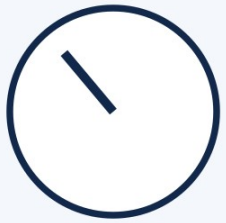
Put the two inputs on a pair of axes and every combination is a point on a plane. A unit with a threshold or sigmoid



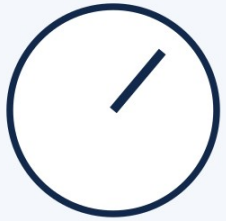
Two Knobs That Slide the Same Line

A shift with no rotation is a change of **intercept** — the baseline firing rate. So raising the threshold and lowering the baseline do the same work. And one straight line is a limited thing.

two knobs

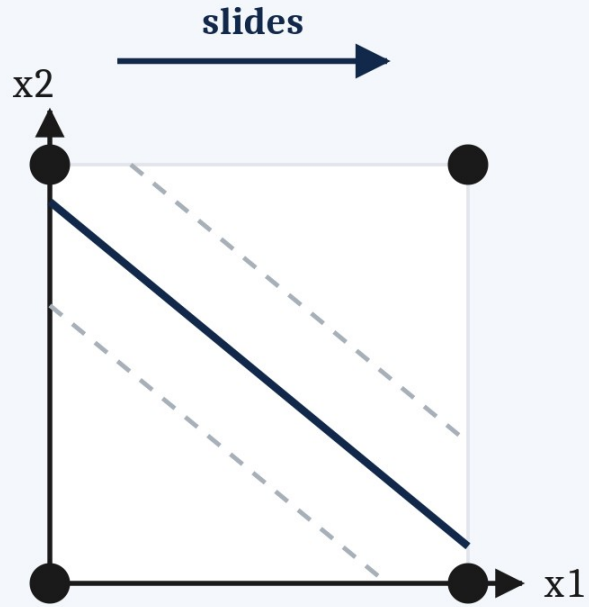


threshold
turn it up



baseline firing rate
turn it down

both slide the line the same way

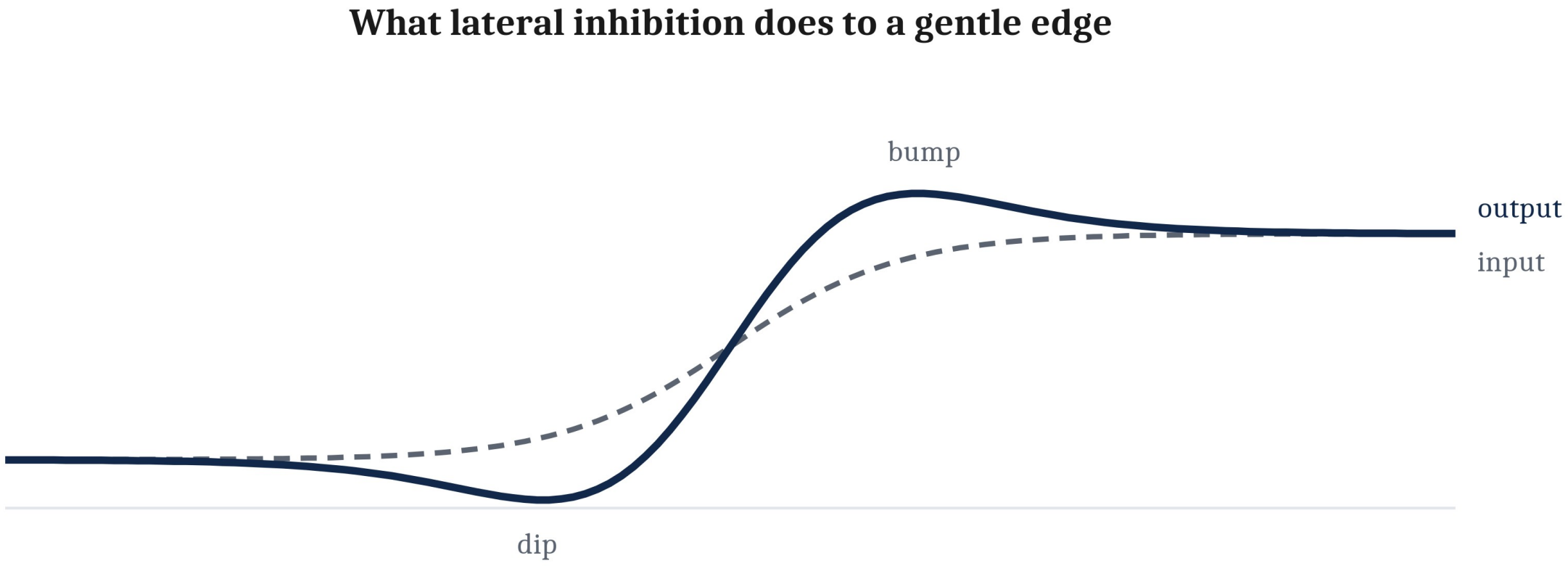


no rotation. only position.

Raising the threshold and lowering the baseline do the same work.

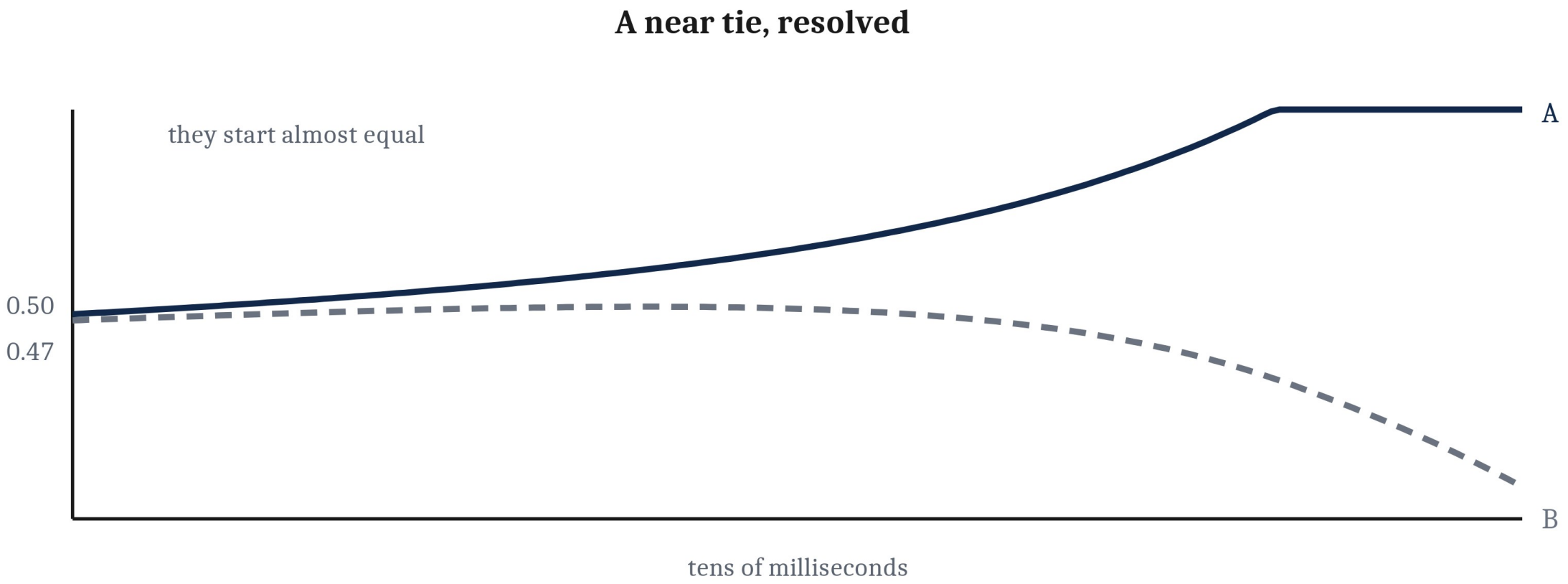
Lateral Inhibition: Reporting the Difference

A row of units, each inhibiting its neighbors in proportion to its own activity. The one that is slightly ahead inhibits harder than it is inhibited, so its lead grows: **a shallow step in, a steep step out**.



Mutual Inhibition, and Choosing Exactly One

Two units, each inhibiting the other. From a near tie the lead runs away, and within a few tens of milliseconds one is firing and the other is silent. **Mutual inhibition** forces **winner-take-all**.



The circuit does not average a near tie. It resolves one.

A State Can Be a Circuit That Is Still Running

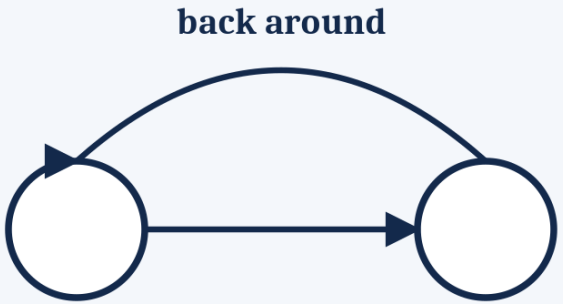
Everything so far ran one way — **feedforward**. A **feedback** or **recurrent connection** runs a unit's output back toward its own input. The stimulus lasts a moment; the activity lasts as long as the loop runs.

feedforward



input to unit to output, one direction

recurrent



every connection in the loop still runs one way

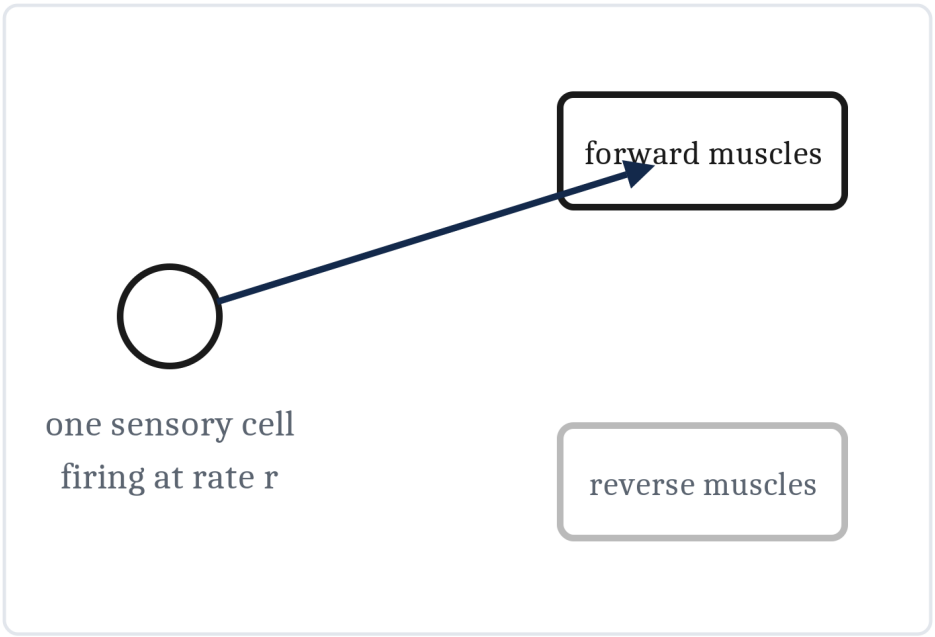


Where Valence Lives

A sensory neuron carries a rate, and nothing else — a food cell and a toxin cell can be identical in every respect a downstream cell could measure. What makes one a good-detector is **what it is wired to**.

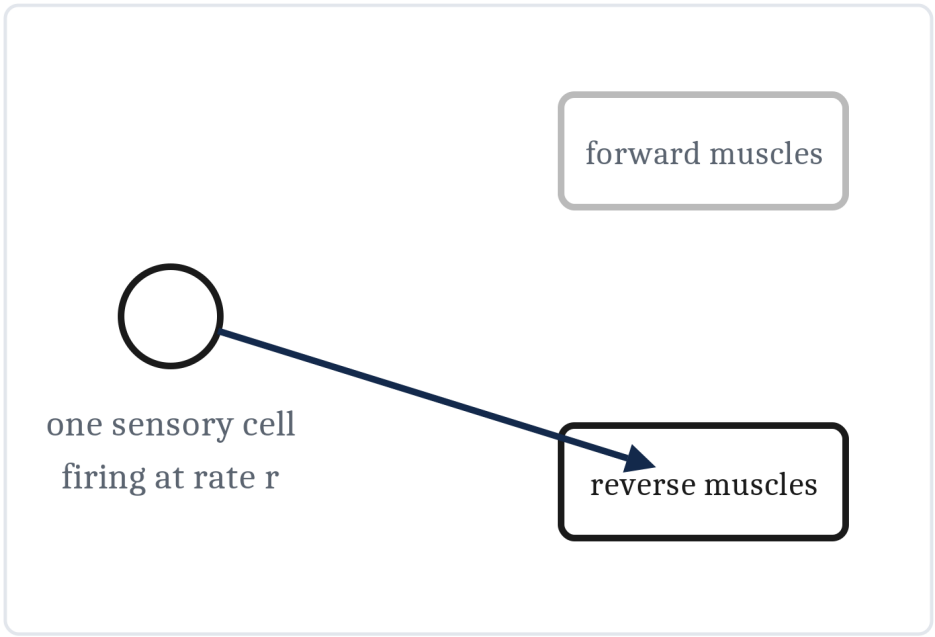
The same cell, wired two ways

Wired one way



the animal moves toward it

Wired the other



the animal moves away from it

Nothing about the cell or its rate differs. Only what it is wired to.

Why a Space, and Not a List

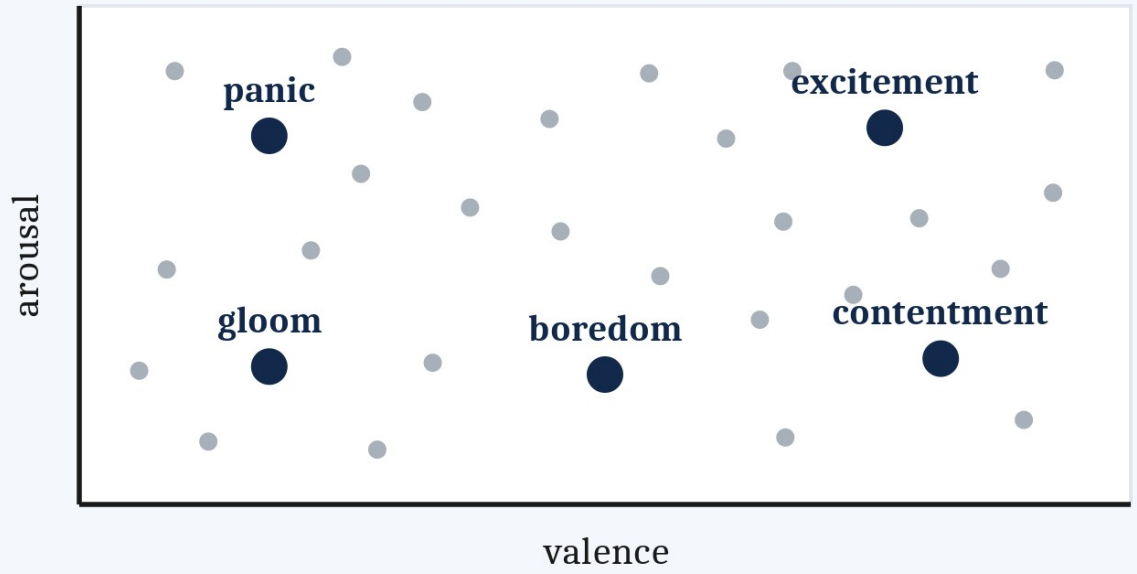
Algorithmically, an affective state is **two numbers** — how good or bad things are, and how activated the animal is — and what it does next is a function of its position. A list holds only what someone thought to name.

a list

- fear
- anger
- joy
- sadness
- disgust
- surprise

written in advance, and holding only what someone thought to name

a space

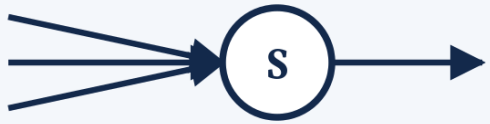


every position is a possible state, including the ones with no word

And two numbers describe a worm and a person in the same terms.

The Same Motifs in Silicon

An artificial network is also built from units that take weighted inputs, sum them, and pass the total through an activation function — and it is wired into the same patterns. One diagram serves both.



weighted inputs, summed, through an activation function -- the same unit in both

convergence and divergence	every layer of a network is made of many-onto-one and one-onto-many
mutual inhibition	wherever a network has to settle on one output among several
a recurrent loop	how a network holds a state across time

One diagram for both -- Module 1's multiple realizability, at the scale of the circuit.

What a Fixed Circuit Cannot Do

Every demand raised in the first lecture now has a wiring pattern that meets it. And every one of those arrangements has **fixed weights** — set in advance, unchanged by anything that happens to the circuit.

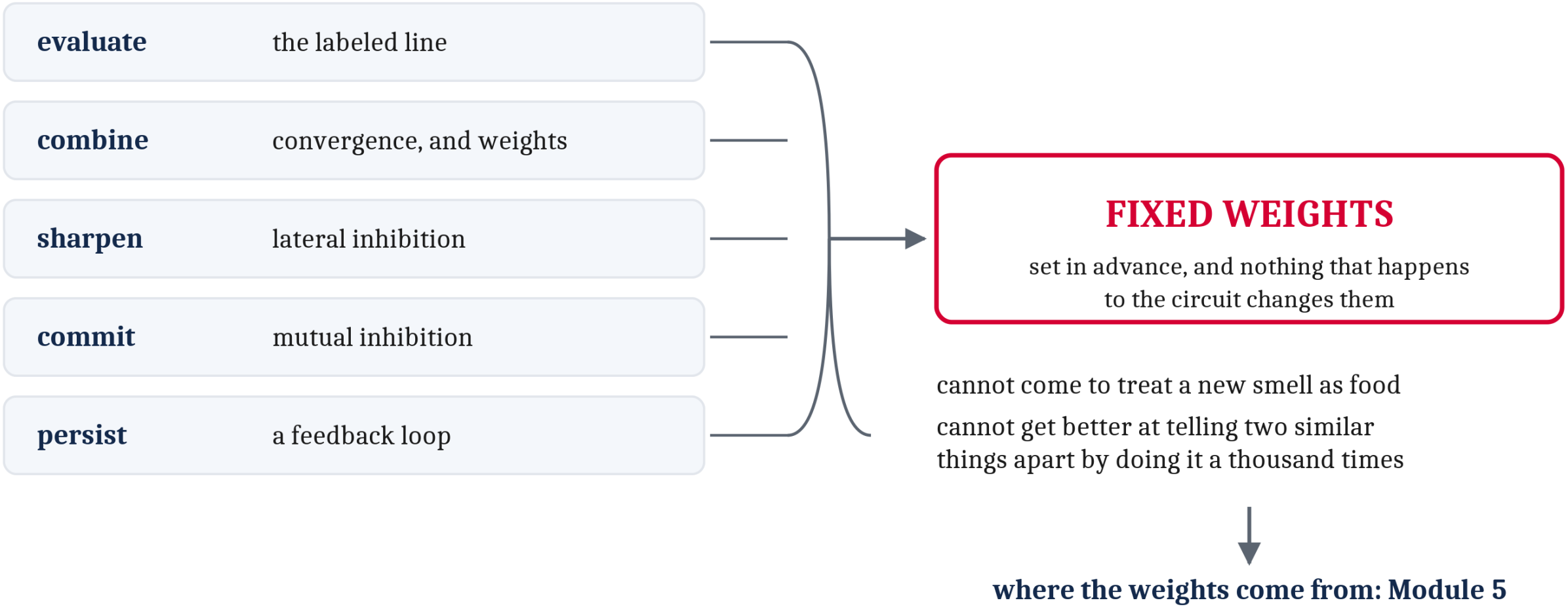


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15	Every demand met, and the one limit they share	fig-fixed-weights.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work